

Assignment No 04

Name: Om Varma

Roll No: 28

PRN No: 1252010028

Question:

Operator overloading with Parameterized constructor for Complex number addition, Subtraction and Multiplication.

Code:

```
#include <iostream>
#include <limits>
using namespace std;

class Complex {
private:
    double real;
    double imag;

public:
    Complex() {
        real = 0;
        imag = 0;
    }

    Complex(double r, double i) {
        real = r;
        imag = i;
    }

    void input() {
        cout << "Enter real part: ";
        while (!(cin >> real)){
            cout << "Invalid input! Enter numeric value: "<<endl;
            cin.clear();
            cin.ignore(numeric_limits<streamsize>::max(), '\n');
        }

        cout << "Enter imaginary part: ";
        while (!(cin >> imag)){
            cout << "Invalid input! Enter numeric value: "<<endl;
            cin.clear();
            cin.ignore(numeric_limits<streamsize>::max(), '\n');
        }
    }

    void display(){
        cout << real;
        if (imag >= 0){
            cout << " + " << imag << "i"<<endl;
        }
    }
};
```

```

    }
    else{
        cout << " - " << -imag << "i"<<endl;
    }
}

Complex operator+(Complex& c){
    return Complex(real + c.real, imag + c.imag);
}

Complex operator-(Complex& c){
    return Complex(real - c.real, imag - c.imag);
}

Complex operator*(Complex& c){
    double r = real * c.real - imag * c.imag;
    double i = real * c.imag + imag * c.real;
    return Complex(r, i);
}
};

```

```

int main() {
    Complex c1, c2;

    cout << "Enter first complex number:"<<endl;
    c1.input();

    cout << "Enter second complex number:"<<endl;
    c2.input();

    Complex sum = c1 + c2;
    Complex diff = c1 - c2;
    Complex prod = c1 * c2;

    cout << "First Complex: "<<endl;
    c1.display();

    cout << "Second Complex: "<<endl;
    c2.display();

    cout << "Addition: "<<endl;
    sum.display();

    cout << "Subtraction: "<<endl;
    diff.display();

    cout << "Multiplication: "<<endl;
    prod.display();

    cout << endl;
    return 0;
}

```

Output:

```
PS D:\STUDY\B.Tech\SEM IV\OOP\Assignment 5> .\complex.exe
Enter first complex number:
Enter real part: 5
Enter imaginary part: 7.6
Enter second complex number:
Enter real part: 2
Enter imaginary part: 4.5
First Complex:
5 + 7.6i
Second Complex:
2 + 4.5i
Addition:
7 + 12.1i
Subtraction:
3 + 3.1i
Multiplication:
-24.2 + 37.7i
```